WORKED SOLUTION

STEP I 2016, Question 11

Mechanics • Projectile and tower

STEP I 2016, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

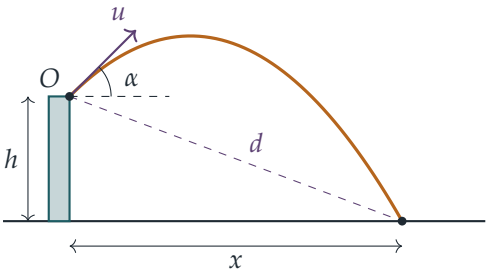
step.maths.org/questions/16-s1-q11

The key idea

Write down the horizontal and vertical motion, eliminate the time, and convert the squares of $\sin\alpha$ and $\cos\alpha$ into double angles. For the greatest range, treat the landing equation as defining x implicitly as a function of α and differentiate it: at the maximum $\mathrm{d}x/\mathrm{d}\alpha = 0$, and the rest is trigonometric algebra in the single unknown $\cos 2\alpha$.

Solution

We model P as a particle moving under gravity alone (no air resistance), with the plain horizontal. Take axes with origin at the base of the tower, x horizontal in the direction of firing and y vertically upwards, so that O is the point $(0, h)$.



The landing equation

At time t after firing,

$$X = ut\cos\alpha, \qquad Y = h + ut\sin\alpha - \tfrac{1}{2}gt^2.$$

The tower stands upright and P lands at some distance $x > 0$, so $\cos\alpha \neq 0$ and $t = \dfrac{x}{u\cos\alpha}$ at landing, where $Y = 0$:

$$0 = h + x\tan\alpha - \frac{gx^2}{2u^2\cos^2\alpha}.$$

Multiply through by $2\cos^2\alpha$ and use $2\cos^2\alpha = 1 + \cos 2\alpha$ and $2\sin\alpha\cos\alpha = \sin 2\alpha$:

$$\frac{gx^2}{u^2} = 2h\cos^2\alpha + 2x\sin\alpha\cos\alpha = h(1+\cos 2\alpha) + x\sin 2\alpha,$$

which is the required equation.

The greatest horizontal distance

The angle α can range over $-\frac{\pi}{2} < \alpha < \frac{\pi}{2}$ (firing downwards is allowed). For each such α the landing distance x is a positive, smoothly varying function of α, and $x \to 0$ as $\alpha \to \pm\frac{\pi}{2}$, because the horizontal speed $u\cos\alpha$ tends to 0 while the time of flight stays bounded. So x has a greatest value at some interior α, and there $\mathrm{d}x/\mathrm{d}\alpha = 0$.

Differentiate the landing equation with respect to α, writing $x' = \mathrm{d}x/\mathrm{d}\alpha$:

$$\frac{2gx}{u^2}\,x' = -2h\sin 2\alpha + x'\sin 2\alpha + 2x\cos 2\alpha.$$

Setting $x' = 0$ leaves

$$x\cos 2\alpha = h\sin 2\alpha.$$

Here $\cos 2\alpha \neq 0$: otherwise $h\sin 2\alpha = 0$ with $h > 0$ would force $\sin 2\alpha = 0$ as well, which is impossible. So at the maximum

$$x = h\tan 2\alpha,$$

as required.

> **Ask yourself** We now have two equations in x and α. How do we get rid of x neatly?

Substitute $x = h\tan 2\alpha$ into the landing equation and write $c = \cos 2\alpha$, $s = \sin 2\alpha$. The right-hand side becomes

$$h(1+c) + h\,\frac{s^2}{c} = \frac{h\,(c + c^2 + s^2)}{c} = \frac{h(1+c)}{c},$$

while the left-hand side is

$$\frac{gh^2}{u^2}\cdot\frac{s^2}{c^2} = \frac{gh^2(1-c)(1+c)}{u^2c^2}.$$

Equating, and dividing by $h(1+c)/c$, which is non-zero since $c \neq -1$ (we showed $c \neq 0$, and $c = -1$ would give $\alpha = \pm\frac{\pi}{2}$):

$$\frac{gh(1-c)}{u^2c} = 1 \quad\Longrightarrow\quad gh - ghc = u^2c \quad\Longrightarrow\quad \cos 2\alpha = \frac{gh}{u^2+gh}.$$

Since this is the only possible value at a stationary point, and $\tan 2\alpha = x/h > 0$ picks out $0 < 2\alpha < \frac{\pi}{2}$, there is exactly one stationary point and it is the maximum. (For the record, $\tan 2\alpha = \dfrac{u\sqrt{u^2+2gh}}{gh}$, so the greatest distance is $x = \dfrac{u\sqrt{u^2+2gh}}{g}$.)

The greatest distance from O

The landing point is at $(x, 0)$, so its distance from $O = (0, h)$ is

$$d = \sqrt{x^2 + h^2}.$$

With h fixed, d increases with x, so d is greatest exactly when x is greatest. At that point $x = h\tan 2\alpha$, so

$$d = \sqrt{h^2\tan^2 2\alpha + h^2} = h\sec 2\alpha = h \cdot \frac{u^2+gh}{gh} = \frac{u^2}{g} + h,$$

using $\sec 2\alpha > 0$. This is the required greatest distance.

Checks

- With $h = 0$ we get $\cos 2\alpha = 0$, so $\alpha = 45°$, and greatest range u^2/g: the familiar results on level ground.
- Take $u = 10$, $g = 10$, $h = 5$ (SI units). Then $\cos 2\alpha = \frac{50}{150} = \frac{1}{3}$, the greatest range is $\sqrt{100 \cdot 200}/10 = 10\sqrt{2} \approx 14.14$, and $d = \sqrt{200 + 25} = 15 = \frac{100}{10} + 5$, as predicted. A computer search over α confirms that 14.14 is the largest landing distance.
- As h grows, $\cos 2\alpha \to 1$, so the best angle tends to 0: from a very tall tower you should fire almost horizontally.

Common slips

- Writing $x = h\tan 2\alpha$ and then forgetting that it holds only at the maximum, not for every α.
- Dividing by $1 + \cos 2\alpha$ or $\cos 2\alpha$ without saying why they are non-zero.
- Maximising d separately instead of noticing that $d^2 = x^2 + h^2$ is maximised at the same angle as x.

Beyond STEP

The answer $d = u^2/g + h$ has a lovely geometric explanation. All the trajectories with speed u from O touch a single *envelope*, the "parabola of safety", which has its focus at O and its directrix a height u^2/g above O. The furthest landing point lies on this envelope, so by the focus–directrix property its distance from O equals its distance

below the directrix, namely $u^2/g + h$. Envelopes of families of curves are a standard topic in first-year calculus and mechanics. Kleppner and Kolenkow, *An Introduction to Mechanics*, is a good next step for projectile problems treated with this kind of care.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.